Cambridge (CIE) A Level Maths: Pure 1

Your notes

Sequences & Series

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Your notes

Language of sequences & series

What is a progression (sequence)?

- A **progression** (or **sequence**) is an **ordered** set of numbers with a rule for finding all the numbers in the progression

e.g. 1, 4, 7, 10, 13, ... IS A SEQUENCE

THE RULE COULD BE GIVEN AS:
"START AT 1. EVERY NUMBER AFTER THAT IS THREE MORE THAN THE NUMBER BEFORE IT"

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- The numbers in a sequence are called **terms**
- The terms of a progression are often referred to by letters with a subscript

SO IN THE SEQUENCE ABOVE

$u_1 = 1 \quad u_2 = 4 \quad u_3 = 7 \quad u_4 = 10 \quad u_5 = 13$

THE GENERAL TERM OF THE SEQUENCE WOULD BE

$u_n = 3n - 2$

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What is a series?

- You get a **series** by summing up the terms in a **progression**

e.g. FOR THE SEQUENCE 1, 4, 7, 10, 13, ...

THE ASSOCIATED SERIES IS $1 + 4 + 7 + 10 + 13 + \ldots$

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- We use the notation S_n to refer to the sum of the first n terms in the progression

  ie. $S_n = u_1 + u_2 + u_3 + \ldots + u_n$

SO FOR THE SERIES ABOVE

$$S_1 = 1$$
$$S_2 = 1+4 = 5$$
$$S_3 = 1+4+7 = 12$$
$$S_4 = 1+4+7+10 = 22$$
$$S_5 = 1+4+7+10+13 = 35$$

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What are increasing, decreasing and periodic progressions?

- A progression is **increasing** if $u_{n+1} > u_n$ for all positive integers n – ie if every term is greater than the term before it
- A progression is **decreasing** if $u_{n+1} < u_n$ for all positive integers n – ie if every term is less than the term before it

e.g.

3, 7, 11, 15, 19, ... IS AN INCREASING SEQUENCE

64, 32, 16, 8, 4, ... IS A DECREASING SEQUENCE

1, −2, 3, −4, 5, ... IS NEITHER AN INCREASING NOR A DECREASING SEQUENCE

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- A progression is **periodic** if the terms repeat in a cycle
- The **order** (or **period**) of a periodic progression is the number of terms in each repeating cycle

e.g.

1, 2, 3, 1, 2, 3, 1, 2, 3, ... IS A PERIODIC SEQUENCE OF ORDER 3

2, −2, 2, −2, 2, −2, ... IS A PERIODIC SEQUENCE OF ORDER 2

0, 0, 0, 0, 0, 0, 0, ... IS A PERIODIC SEQUENCE OF ORDER 1

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Examiner Tips and Tricks

Look out for progressions defined by trigonometric functions – this can be a way of 'hiding' a periodic function.

$u_n = \cos(180n^\circ)$

$u_1 = \cos(180^\circ) = -1$
$u_2 = \cos(360^\circ) = 1$
$u_3 = \cos(540^\circ) = -1$
$u_4 = \cos(720^\circ) = 1$

THIS IS THE SEQUENCE $-1, 1, -1, 1, -1, 1, \ldots$
(i.e. A PERIODIC SEQUENCE OF ORDER 2)

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Worked Example

Determine the order of the following periodic functions

a) $u_n = 3 + (-1)^n$

b) $u_n = \sin(90n^\circ)$

WORK OUT THE TERMS AND LOOK FOR A PATTERN

a) $u_1 = 3 + (-1)^1 = 3 - 1 = 2$
$u_2 = 3 + (-1)^2 = 3 + 1 = 4$
$u_3 = 3 + (-1)^3 = 3 - 1 = 2$
$u_4 = 3 + (-1)^4 = 3 + 1 = 4$

THE SEQUENCE IS $2, 4, 2, 4, 2, 4, \ldots$

PERIODIC WITH ORDER 2

b) $u_1 = \sin 90^\circ = 1$ $\quad u_2 = \sin 180^\circ = 0$
$u_3 = \sin 270^\circ = -1$ $\quad u_4 = \sin 360^\circ = 0$
$u_5 = \sin 450^\circ = 1$ $\quad u_6 = \sin 540^\circ = 0$
$u_7 = \sin 630^\circ = -1$ $\quad u_8 = \sin 720^\circ = 0$

SEQUENCE IS $1, 0, -1, 0, 1, 0, -1, 0, \ldots$

PERIODIC WITH ORDER 4

BE CAREFUL HERE. EVERY SECOND TERM IS A ZERO, BUT IT TAKES FOUR TERMS FOR THE PATTERN TO REPEAT

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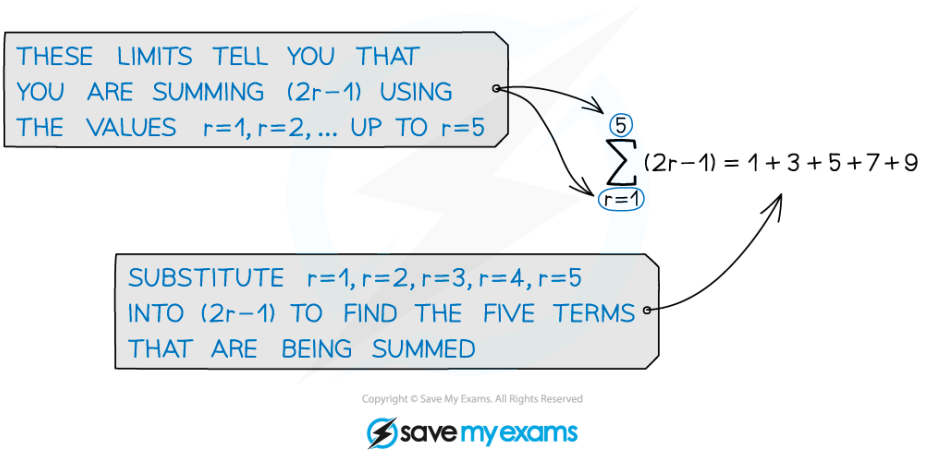
Sigma notation

What is sigma notation?

- The symbol Σ is the capital Greek letter sigma – that's why it's called 'sigma notation'!
- 'Σ' stands for 'sum' – the expression to the right of the Σ tells you what is being summed, and the limits above and below tell you which terms you are summing



- Be careful – the limits don't have to start with 1!
  - For example: $\sum_{r=0}^{4}(2r+1)$ or $\sum_{r=7}^{11}(2r-13)$

How do I use sigma notation?

- Sigma notation can be used to represent both **arithmetic progressions** and **geometric progressions**
  - **Arithmetic** will have the form $A + Br$
  - **Geometric** will have the form $A \times B^r$
  - Writing out the first few terms will help you

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$$\sum_{r=1}^{5}(5r+1) = 6 + 11 + 16 + 21 + 26$$

THIS IS THE SUM OF THE FIRST FIVE TERMS OF THE ARITHMETIC SERIES WITH FIRST TERM 6 AND COMMON DIFFERENCE 5

$$\sum_{r=1}^{5}(4\times 3^{r}) = 12 + 36 + 108 + 324 + 972$$

THIS IS THE SUM OF THE FIRST FIVE TERMS OF THE GEOMETRIC SERIES WITH FIRST TERM 12 AND COMMON RATIO 3

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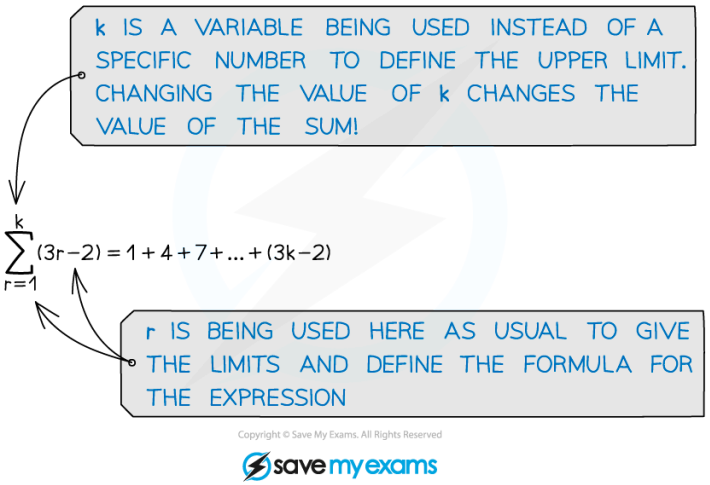
- To work out such a sum use the formulae for the sum of arithmetic and geometric progressions
- As long as the expressions being summed are the same you can add and subtract in sigma notation
  - For example:

$$\sum_{r=1}^{6}(4r+7) + \sum_{r=7}^{11}(4r+7) = \sum_{r=1}^{11}(4r+7)$$

$$\sum_{r=1}^{100}(7\times 2^{r}) - \sum_{r=1}^{50}(7\times 2^{r}) = \sum_{r=51}^{100}(7\times 2^{r})$$

Examiner Tips and Tricks

Be careful when more than one letter appears in a sigma notation question.



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Worked Example

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Evaluate:

a) $\sum_{r=50}^{99} (2r+1)$

b) $\sum_{r=k-4}^{12} (3 \times 4^r)$ when $k = 7$

"EVALUATE" HERE MEANS "FIND THE VALUE OF"

a) THIS IS THE SUM $101 + 103 + 105 + \ldots + 199$
ARITHMETIC SERIES WITH $a=101$, $d=2$, AND LAST TERM $l=199$

$S_{50} = \frac{50}{2}(101+199) = 7500$

$S_n = \frac{n}{2}(a+l)$

BE CAREFUL HERE! THERE ARE 50 NUMBERS (NOT 49) FROM $r=50$ TO $r=99$. SO $n=50$

b) FOR $k=7$ THIS IS

$$\sum_{r=3}^{12}(3\times4^r) = 3\times4^3 + 3\times4^4 + 3\times4^5 + \ldots + 3\times4^{12}$$
$$= 3\times4^3 + 4(3\times4^3) + 4^2(3\times4^3) + \ldots + 4^9(3\times4^3)$$
$$= 192 + 4(192) + 4^2(192) + \ldots + 4^9(192)$$

THAT IS THE FIRST 10 TERMS OF THE GEOMETRIC SERIES WITH $a=192$ AND $r=4$

$S_{10} = \frac{192(4^{10}-1)}{4-1} = 67\,108\,800$

$S_n = \frac{a(r^n-1)}{r-1}$

BE CAREFUL! THIS r IS THE COMMON RATIO OF THE GEOMETRIC SERIES. IT IS DIFFERENT FROM THE r IN THE SIGMA NOTATION.

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Recurrence relations

What is a recurrence relation?

- A **recurrence relation** describes each term in a progression as a function of the previous term – ie $\boldsymbol{u_{n+1} = f(u_n)}$
- Along with the first term of the sequence, this allows you to generate the sequence term by term

e.g. FIND THE FIRST FOUR TERMS OF THE SEQUENCE DEFINED BY $u_{n+1} = u_n + 3,\ u_1 = 2$

$u_1 = 2 \quad u_2 = u_1 + 3 = 2 + 3 = 5$

$u_3 = u_2 + 3 = 5 + 3 = 8$

$u_4 = u_3 + 3 = 8 + 3 = 11$

THE FIRST FOUR TERMS ARE 2, 5, 8, 11

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- Both **arithmetic progressions** and **geometric progressions** can be defined using recurrence relations
  - **Arithmetic** can be defined by $u_{n+1} = u_n + d,\ u_1 = a$
  - **Geometric** can be defined by $u_{n+1} = u_n \times r,\ u_1 = a$

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$$u_{n+1} = u_n - 5, \quad u_1 = 9$$

THIS IS THE SEQUENCE $9, 4, -1, -6, -11, \ldots$
(i.e. ARITHMETIC SEQUENCE WITH FIRST TERM 9 AND COMMON DIFFERENCE -5)

$$u_{n+1} = 2u_n, \quad u_1 = 3$$

THIS IS THE SEQUENCE $3, 6, 12, 24, 48, \ldots$
(i.e. GEOMETRIC SEQUENCE WITH FIRST TERM 3 AND COMMON RATIO 2)

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- However, you can also define progressions that are neither arithmetic nor geometric

$$u_{n+1} = 2u_n + 3, \quad u_1 = 1$$

THIS IS THE SEQUENCE $1, 5, 13, 29, 61, \ldots$
(DON'T CONFUSE THIS WITH ARITHMETIC SEQUENCE $5, 7, 9, 11, \ldots$ DEFINED BY $u_n = 2n + 3$)

$$u_{n+1} = (u_n)^2, \quad u_1 = 2$$

THIS IS THE SEQUENCE $2, 4, 16, 256, 65\,536, \ldots$

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Examiner Tips and Tricks

- For arithmetic or geometric progressions defined by recurrence relations, you can sum the terms using the **arithmetic series** and **geometric** formulae.
- To sum up the terms of other progressions, you may have to think about the series and find a clever trick.

Worked Example

Your notes

A sequence $u_1,\ u_2,\ u_3,\ \ldots$ is defined by

$$u_1 = 2$$

$$u_{n+1} = u_n^2 - 3$$

Find $\sum_{r=1}^{101} u_r$

$u_1 = 2 \qquad u_2 = 2^2 - 3 = 1 \qquad u_3 = 1^2 - 3 = -2$

$u_4 = (-2)^2 - 3 = 1 \qquad u_5 = 1^2 - 3 = -2$

WORK OUT THE TERMS AND LOOK FOR A PATTERN

SO THE SEQUENCE IS

$2, 1, -2, 1, -2, 1, -2, \ldots$

IN THE FIRST 101 TERMS THERE WILL BE

ONE 2 $\qquad 1 \times 2 = 2$

FIFTY 1's $\qquad 50 \times 1 = 50$

FIFTY −2's $\qquad 50 \times (-2) = -100$

$2 + 50 + (-100) = -48$

SO $\sum_{r=1}^{101} u_r = -48$

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Modelling with sequences & series

How do I model with sequences and series?

- Many real-life situations can be modelled using progressions
- If a quantity is changing repeatedly by having a fixed amount **added** to or **subtracted** from it then the use of **arithmetic progressions** is appropriate

e.g. IN JANUARY VINCENT PUTS £100 INTO HIS SAVINGS ACCOUNT. FOR THE REST OF THE YEAR, IN EACH MONTH HE PUTS IN £10 MORE THAN THE MONTH BEFORE.
HOW MUCH WILL HE HAVE PUT INTO HIS SAVINGS ACCOUNT BY THE END OF THE YEAR?

THE TOTAL AMOUNT HE PUTS INTO HIS SAVINGS ACCOUNT IS:

$$100 + 110 + 120 + 130 + \ldots$$

THAT IS AN ARITHMETIC SERIES WITH FIRST TERM $a = 100$ AND COMMON DIFFERENCE $d = 10$. THERE WILL BE 12 TERMS IN THE SUM SO THE TOTAL IS:

$$S_{12} = \frac{12}{2}(2(100) + (12-1)10) = £1860$$

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- If a quantity is changing repeatedly by a fixed **percentage**, or by being **multiplied** repeatedly by a fixed amount, then the use of **geometric progressions** is appropriate

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e.g. LOWENA'S BUSINESS MAKES £20,000 PROFIT IN ITS FIRST YEAR. SHE EXPECTS THAT DURING THE FIRST TEN YEARS THE BUSINESS' PROFIT WILL INCREASE BY 10% EACH YEAR. IF SHE IS CORRECT THEN WHAT WILL THE TOTAL PROFIT BE OVER THOSE TEN YEARS, TO THE NEAREST THOUSAND POUNDS?

THE TOTAL PROFIT WILL BE:

$$20\,000 + (1.1 \times 20\,000) + (1.21 \times 20\,000) + \ldots$$

THAT IS A GEOMETRIC SERIES WITH FIRST TERM $a = 20\,000$ AND COMMON RATIO $d = 1.1$. THERE WILL BE 10 TERMS IN THE SUM SO THE TOTAL IS:

$$S_{10} = \frac{20\,000\,(1.1^{10} - 1)}{(1.1 - 1)} = 318\,748.492\ldots$$

$$= £319\,000 \text{ TO THE NEAREST } £1000$$

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Examiner Tips and Tricks

- Remember, an exam question won't always tell you to use progression methods, so you've got to be able to spot 'hidden' progression questions.
- To help you do this, be suspicious of questions about savings accounts, salaries, sales commissions, profits and the like – these are often progression questions in disguise!

Worked Example

Your notes

Anna's grandfather gives her £10 on her first birthday. He says that he will give her £20 on her second birthday, £40 on her third birthday, and so on, doubling his gift each year until she is 18. If he goes through with his plan, then

a) How much will he have to give Anna on her eighteenth birthday?

b) How much in total will he have given her by (and including) her eighteenth birthday?

a) THIS IS A GEOMETRIC SEQUENCE WITH $a = 10 \quad r = 2$

THE EIGHTEENTH TERM IS

$10 \times 2^{17} = £1\,310\,720$ ← ar^{n-1}

b) THIS IS A GEOMETRIC SERIES WITH $a = 10 \quad r = 2$

$S_{18} = \frac{10(2^{18}-1)}{2-1} = £2\,621\,430$ ← $S_n = \frac{a(r^n-1)}{r-1}$

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